AJEENKYA D Y PATIL UNIVERSITY — SCHOOL OF ENGINEERING

BTECH (CTIS) | Sem VII — Research Methodology (Continuous Assignment-1)

Name of Assignment	Continuous Assignment-1 (CA-1)
Course Name	Research Methodology (BTECH CTIS | Sem VII)
Faculty Name	Prof. Parmeshwari Aland (Asst. Professor / HOD-BCA)
Date / Deadline	September 2026 / October 2026
Problem Statement (SIH 26160)	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (NTRO)
Student Name & Group	**Pratik Shivekar** (Group: Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar)
Individual Focus Area	LLM-Driven Protocol State Serialization, Active/Passive Vulnerability Assessment, and Automated Risk Remediation

1. Task 1 (LO1): Definition and Conceptualisation of Research and Its Purpose

Research is a systematic, rigorous, and objective process of scientific investigation designed to uncover novel principles, validate theoretical models against empirical observations, and engineer verifiable solutions to complex problems. It differs fundamentally from routine software development or ad-hoc network debugging because it operates within a formal methodological framework—defining explicit research questions, synthesizing peer-reviewed literature to locate knowledge gaps, formulating testable hypotheses, and evaluating outcomes through reproducible quantitative metrics.

In Cybersecurity and Automated Protocol Analysis, research focuses on the intersection of formal methods, network systems engineering, and artificial intelligence. Cryptographic protocols such as IPsec and IKEv1/IKEv2 are governed by intricate cross-message state dependencies and strict RFC security constraints. Deviations between protocol specifications and real-world configurations or implementations create exploitable attack surfaces, including state-machine bypasses, amplification denial-of-service (DoS), and side-channel leaks. The purpose of this research is to advance automated security assessment through: (1) Analytical Inquiry into multi-layer protocol vulnerabilities (classical, implementation-level, and post-quantum); (2) Methodological Innovation combining Large Language Models (LLMs) and machine learning to parse protocol states and encrypted flows; and (3) Prescriptive Engineering to automatically generate actionable, prioritized security recommendations.

2. Tasks 2 & 3 (LO2 & LO3): Research Area Selection and Topic Justification (300+ Words)

Broad Research Area: Automated Network Protocol Security Assessment, Large Language Models (LLMs) for Cybersecurity, and Cryptographic Resilience.
Specific Research Topic: AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework (Smart India Hackathon 2026 Problem Statement ID: 26160, National Technical Research Organisation - NTRO).

Justification of Relevance and Feasibility: Enterprise, defense, and critical infrastructure networks rely heavily on IPsec VPNs to guarantee confidentiality, integrity, and authentication across untrusted links. However, securing an IPsec deployment requires configuring dozens of interdependent parameters across IKEv1/IKEv2 key exchange, authentication (PSK, PKI, Kerberos GSSAPI), ESP/AH cipher suites, anti-replay windows, and Tunnel/Transport modes. Research demonstrates that even when cryptographic primitives are mathematically sound, implementation flaws, misconfigured anti-replay windows, or deterministic traffic patterns allow adversaries to mount stealth MITM DoS amplification attacks (Herzberg & Shulman, 2009), selective delay attacks on critical time synchronization (Annessi et al., 2018), and specification-violating state transitions (Chen et al., 2026).

Traditional packet analysis tools provide raw hexadecimal and field-level dissection but leave the burden of vulnerability reasoning, cross-message state validation, and remediation planning entirely to human experts. Selecting this research topic is highly relevant because modern AI—combining machine learning flow classifiers with LLM-driven specification analysis (such as SeriCrypt)—can automate expert-level interpretation, quantify multi-vector security risks, and produce concrete configuration patches (e.g., strongSwan `swanctl.conf` or Cisco IOS templates). The research is completely feasible using standard PCAP traces, open-source IPsec implementations, and local/API-driven AI inference pipelines.

3. Task 4 (LO4): Design of Research Title with Detailed Explanation (200+ Words)

Designed Research Title: "AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework: LLM-Assisted Protocol State Auditing and Automated Risk Remediation"

Title Explanation and Scope Alignment: The research title is crafted to concisely reflect the problem domain, intelligent methodology, and practical security output of Pratik Shivekar's research focus:
1. 'AI-Powered': Encompasses the hybrid artificial intelligence architecture combining supervised ML classifiers for traffic behavior with Large Language Models (LLMs) for RFC specification parsing and remediation generation.
2. 'IPsec VPN Protocol Analyzer': Defines the exact protocol target—inspecting IKEv1/IKEv2 handshakes, GSSAPI/PKI authentication exchanges, and ESP/AH data tunnels.
3. 'Security Assessment Framework': Signifies a structured evaluation engine that grades deployments across confidentiality, integrity, authentication strength, DoS resilience, and post-quantum readiness.
4. 'LLM-Assisted Protocol State Auditing and Automated Risk Remediation': Highlights Pratik Shivekar's specialized contribution—leveraging context-aware protocol state modeling (inspired by SeriCrypt) and vulnerability taxonomies to detect state-machine deviations, anti-replay misconfigurations, and deprecated primitives, and translating those findings into actionable remediation playbooks.

4.1 Task 8 (LO8): Abstract Draft (Kept Same Across Group)

Virtual Private Networks (VPNs) serve as the foundational backbone for confidential, integrity-protected, and authenticated communication across untrusted public networks. Among existing tunneling architectures, Internet Protocol Security (IPsec) is extensively deployed across enterprise, government, defense, and cloud infrastructures. However, the real-world security posture of an IPsec deployment is highly sensitive to its negotiated cryptographic algorithms, Internet Key Exchange (IKEv1/IKEv2) state transitions, authentication mechanisms, Encapsulating Security Payload (ESP) parameters, operational modes (Tunnel versus Transport), and resilience against emerging post-quantum and side-channel threats. Misconfigurations, deprecated cipher suites, weak Diffie-Hellman groups, improper key management, or protocol implementation flaws frequently expose critical networks to downgrade, man-in-the-middle (MITM), and traffic fingerprinting attacks. Traditional packet analyzers such as Wireshark provide low-level packet visibility but require manual expert interpretation. This research proposes an AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework that combines automated IKE/ESP state-machine parsing, multivariate encrypted traffic metadata analysis, and Large Language Model (LLM)-assisted vulnerability reasoning to automatically fingerprint IPsec deployments, quantify cryptographic and operational risks, and generate actionable, standards-compliant remediation recommendations.

4.2 Task 8 (LO8): Research Introduction (400–500 Words)

Cryptographic network protocols—led by Internet Protocol Security (IPsec)—form the foundational trust layer of enterprise wide-area networks, government communications, and critical national infrastructures. By operating at the network layer (OSI Layer 3), IPsec provides transparent confidentiality, data origin authentication, connectionless integrity, and anti-replay protection for all upper-layer TCP/UDP traffic (Elliott et al., 2003; Mink et al., 2010; Baseri et al., 2024). However, the security assurance of an IPsec VPN deployment is not guaranteed merely by enabling encryption. Rather, it is governed by whether the negotiated cryptographic algorithms, key exchange protocols (IKEv1/IKEv2), authentication bindings (PKI, Pre-Shared Keys, or Kerberos V5 GSSAPI), operational modes (Tunnel or Transport), and anti-replay window parameters strictly adhere to modern security standards.

Empirical cybersecurity research has repeatedly shown that subtle protocol misconfigurations and implementation deviations can catastrophically weaken IPsec tunnels. For example, improperly sized or disabled IPsec anti-replay windows allow stealth Man-in-the-Middle (MITM) adversaries to launch amplifying TCP degradation-of-service (DoS) attacks by duplicating and reordering a minimal number of ESP packets (Herzberg & Shulman, 2009). Similarly, statistical traffic analysis of packet lengths and inter-arrival times can identify encapsulated Precision Time Protocol (PTP) messages or user OS/browser profiles inside IPsec tunnels, enabling selective delay attacks and privacy breaches even when strong encryption is active (Dvir et al., 2016; Annessi et al., 2018). Furthermore, deviations between RFC natural-language specifications and C/C++ daemon implementations introduce state-machine bypass vulnerabilities in IKEv1 and IKEv2 (Chen et al., 2026).

While traditional packet analyzers such as Wireshark dissect individual packet headers, they lack contextual intelligence to validate cross-message cryptographic dependencies, evaluate multi-layer quantum and classical risks, or prescribe configuration fixes. To overcome these limitations, this paper proposes an AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. Integrating ML-based encrypted traffic analysis, context-aware IKE/ESP state-machine auditing, and LLM-driven remediation synthesis, the framework provides automated, end-to-end security assessment for modern IPsec deployments.

5. Task 5 (LO5): Preliminary Findings and Literature Review Summary (500+ Words)

A comprehensive review of 25 peer-reviewed research papers was carried out to establish the state of the art in IPsec protocol security, encrypted traffic analysis, and intelligent vulnerability assessment. Within the three-member research group, Pratik Shivekar focused primarily on Papers 18 through 25—covering LLM-driven cryptographic protocol serialization, adversarial ML traffic classification, stealth MITM DoS and delay attacks on IPsec, authentication integration, and multi-layer quantum vulnerability taxonomies—while synthesizing insights across all 25 papers.

1. LLM-Driven Cryptographic Protocol State Analysis and Fuzzing: Chen et al. (2026) [Paper 18] introduced SeriCrypt, a pioneering LLM-driven, context-aware serialization framework for cryptographic protocols including IKEv1/IKEv2, TLS 1.2/1.3, and SSH. By using an LLM to extract field constraints, cross-message state dependencies, and cryptographic computation rules from unstructured RFC documents into a Cryptographic Domain-Specific Language (CDSL) and Protocol Abstract Syntax Tree (PAST), SeriCrypt uncovered five specification violations and achieved significantly deeper state coverage in IKEv1/v2 testing than traditional fuzzers.

2. Active and Passive Attacks on Encrypted IPsec Channels (DoS, Delay, and Fingerprinting): A critical insight from the literature is that encryption alone does not guarantee channel resilience. Herzberg and Shulman (2009) [Paper 20] formally modeled stealth MITM adversaries capable of launching amplifying Denial/Degradation of Service (DoS) attacks on TCP over IPsec ESP, proving that missing or undersized anti-replay windows lead to severe throughput collapse. Annessi et al. (2018) [Paper 21] demonstrated that statistical traffic analysis of packet lengths and timing identifies Precision Time Protocol (PTP) messages inside IPsec tunnels, enabling selective delay attacks on critical infrastructure clocks. Likewise, Dvir et al. (2016) [Paper 19] showed that robust machine learning classifiers can infer victim OS, browser, and application tuples with >85% accuracy despite VPN tunneling and cipher parameter changes, aligning with recent deep learning advances in GETA (Gunasekara et al., 2026) [Paper 15] and HW-NAS 1D-CNNs (Chehade et al., 2025) [Paper 17].

3. Authentication Mechanisms and Quantum-Safe IPsec Architectures: Song-Ho et al. (2016) [Paper 23] addressed enterprise authentication by integrating public-key Kerberos V5 (`PKINIT_AS` with ECDH/ECSig) and `gssapi_krb` into IPsec VPNs, highlighting how sensitive GSSAPI-IPsec parameter bindings are to misconfiguration. On the quantum security front, foundational works by Elliott et al. (2003) [Paper 22] on the DARPA Quantum Network and Mink et al. (2010) [Paper 25] at NIST established the architectural blueprints for coupling high-speed QKD with IPsec SAs. More recently, Baseri et al. (2024) [Paper 24] (published in Elsevier *Computers & Security*) provided an exhaustive vulnerability assessment of TCP/IP protocols—including IPsec AH, ESP, and IKEv2—across quantum migration stages, complementing the empirical hybrid PQC/QKD evaluations of Blanco-Romero et al. (2025) [Paper 3], Alam (2026) [Paper 4], and Mallick et al. (2026) [Paper 9].

Identified Research Gap: While prior studies separately demonstrate LLM-based RFC parsing for active fuzzing (SeriCrypt), ML-based passive flow classification (GETA), and theoretical vulnerability matrices (Baseri et al., 2024), there is no integrated AI framework that passively and actively audits live IPsec VPN deployments, evaluates both cryptographic and flow-level DoS/side-channel risks, and automatically synthesizes vendor-specific remediation configurations.

6. Task 6 (LO6): Data Gathering, Organisation, and Proposed Methodology

In fulfillment of LO6 and Tasks 6–8, 25 peer-reviewed research papers were downloaded as full-text PDFs and cataloged in Reference_Papers_25_CA1.xlsx. Within the group of 3 (Sarvesh Bhamare, Aarohi Londhe, Pratik Shivekar), Pratik Shivekar analyzed Papers 18–25. The proposed AI-Powered IPsec Analyzer comprises four stages: (1) Multi-Source PCAP & Config Ingestion; (2) Deterministic IKEv1/IKEv2 State & Cryptographic Proposal Parser; (3) Multivariate Time-Series Deep Learning Encrypted ESP Flow & Side-Channel Classifier; and (4) LLM-Driven Quantitative Security Risk Scoring and Automated Remediation Synthesis.

7. Task 7 (LO7): Comprehensive Research Paper Outline (500+ Words)

Section 1.0: Abstract & Executive Summary — Scope, AI methodology, and key security assessment metrics.
Section 2.0: Introduction & Problem Statement — IPsec architecture (IKEv1/v2, ESP, AH), Tunnel vs. Transport mode risks, DPI blindness, NTRO Problem Statement 26160, and contributions of Pratik Shivekar.
Section 3.0: Literature Review & Threat Taxonomy — Critical synthesis of 25 papers covering VPN ecosystem scans, PQC/QKD migration (RFC 9242/9370, ML-KEM, XMSS), side-channel/DoS attacks (TPM-FAIL, Stealth MITM), and AI/LLM protocol analysis (GETA, HW-NAS 1D-CNN, SeriCrypt).
Section 4.0: System Architecture — IKEv1/v2 Handshake Dissector, Multivariate Encrypted Flow Classifier, and CDSL State Validator.

Section 5.0: Quantitative Security Risk Scoring & LLM Remediation — Multi-factor scoring across Confidentiality, Integrity, Authentication, Key Exchange, DoS/Side-Channel resilience, and Post-Quantum readiness.
Section 6.0: Experimental Evaluation — Multi-vendor IPsec testbed (strongSwan, Libreswan, OpenBSD iked, P4 BMv2) and ISCXVPN2016 benchmarks.
Section 7.0: Discussion & Viva Voce Insights — Operational scalability, adversarial traffic morphing defenses, and privacy preservation.
Section 8.0: Conclusion, Future Scope & APA References — Summary of contributions and 25+ APA citations.

8. Pratik Shivekar's Assigned Research Papers (Papers 18–25): Reference, Verbatim Abstract, Introduction & Literature Review

Paper 18 [Pratik Shivekar]: SeriCrypt: An LLM-Driven Context-Aware Serialization Framework for Cryptographic Protocols

1. APA Reference: Chen, M., Chen, X., Ju, M., Zhao, D., & Gu, C. (2026). SeriCrypt: An LLM-Driven Context-Aware Serialization Framework for Cryptographic Protocols. arXiv preprint arXiv:2608.24498. https://arxiv.org/abs/2608.24498 (Local File: Paper_18_2608.24498_SeriCrypt_An_LLM_Driven_Context_Aware_Serializatio.pdf)

2. Abstract (Verbatim / Kept Same): Constructing syntactically correct and cryptographically valid message sequences is essential for protocol state machine learning, conformance testing, and fuzzing. Unlike plaintext protocols, cryptographic protocols involve complex cross-message state dependencies and cryptographic computation constraints. Existing automated approaches predominantly target text-based or plaintext protocols, leaving cryptographic message construction largely manual. We present SeriCrypt, an LLM-driven, context-aware serialization framework for cryptographic protocols. It employs a large language model to extract field constraints, state dependencies, and cryptographic computation rules from unstructured protocol specifications into a unified structured intermediate representation, formally characterized by a domain-specific language for cryptographic protocols (CDSL). A protocol-agnostic execution engine parses CDSL declarations, automating field value resolution, cryptographic primitive invocation, and byte-stream serialization. As case studies in protocol security testing, we use the framework to construct violation messages targeting specification-defined security constraints and to support protocol fuzzing, evaluating it on mainstream implementations of TLS 1.2/1.3, IKEv1/v2, SSH, and TLCP. SeriCrypt generated message sequences accepted by all evaluated implementations and completed handshakes in every scenario. Security constraint testing revealed five specification violations, and fuzzing reached deeper protocol states with higher code coverage than mainstream fuzzers under the same time budget, demonstrating the framework's practical value for cryptographic protocol security testing.

3. Introduction: Cryptographic network protocols (e.g., TLS, IPsec, SSH) are the foundational infrastructure of modern network communication security, providing core functions such as confidentiality protection, identity authentication, and integrity verification. The security of these protocols depends not only on the completeness of the specification design but also on whether their implementations strictly adhere to the security constraints defined in the RFCs. Studies show that even when the specification itself is thoroughly verified, deviations at the implementation level can still lead to severe security vulnerabilities such as state machine bypass [1]–[4]. Therefore, security testing against protocol implementations is an indispensable means of ensuring communication security [5]. For encrypted protocols, constructing messages that conform to cryptographic specifications can effectively improve the efficiency of security testing methods such as fuzzing [6], state-machine learning [2], and conformance testing. This requires that messages not only satisfy precise byte-level encoding but also handle complex cryptographic computations within a complete session context, and that their values hold deep dependencies on the states of preceding interactions. This strict requirement of cross-message state consistency and cryptographic correctness poses challenges for automating the construction of such messages. The knowledge required to construct such messages is described in natural language in protocol specifications (e.g., IETF RFCs and national standards). LLMs have natural language understanding capabilities and can automatically extract this protocol knowledge from the specification text, but two problems remain in their practical application: First, directly employing LLMs to gene

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR), 2026 | Methodology: LLM-driven context-aware serialization framework (SeriCrypt) using a Cryptographic Domain-Specific Language (CDSL) to parse RFCs and fuzz IKEv1/IKEv2, TLS, and SSH. | Key Findings: Automatically extracts protocol state dependencies from RFC specifications, uncovering 5 specification violations and achieving superior code coverage in IKEv1/v2 fuzzing. | Research Gap: Focuses on active fuzzing message generation; can be combined with passive ML flow analysis for a unified non-intrusive + active IPsec security assessment platform.] II. PROBLEMDEFINITION ANDDESIGNREQUIREMENTS The generation of cryptographic protocol messages is fundamentally different from that of plaintext protocols. For cryptographic protocols, even if the precise syntactic structure of protocol messages is known, if one cannot access and process the contextual information and cryptographic computations in the session process, it is still impossible to construct a message that can pass peer verification [10], [11]. The complexity is mainly reflected in two aspects: crossmessage context references—some fields need to reference random numbers, negotiated parameters, and ephemeral keys exchanged or negotiated in previous messages; cryptographic computation constraints—some fields are themselves the results of cryptographic operations, such as key derivation, message authentication codes, and digital signatures. Take the Finished message in TLS 1.3 as an example [12]. The syntactic structure of this message is simple, but the construction of its verify data field must take as input the transcript-hash

accumulated from previous handshake messages, combine it with the finished-key derived from the shared secret, and then compute the HMAC. Relying solely on static field format definitions cannot satisfy such dynamic computation requirements. Accordingly, this paper decomposes the problem of automated construction of cryptographic protocol messages into two sub-problems: •Representation problem— a formal language is needed to describe protocol message structures, cross-message state dependencies, and cryptographic computation logic, in order

Paper 19 [Pratik Shivekar]: Robust Machine Learning for Encrypted Traffic Classification

1. APA Reference: Dvir, A., Zion, Y., Muehlstein, J., Pele, O., Hajaj, C., & Dubin, R. (2016). Robust Machine Learning for Encrypted Traffic Classification. arXiv preprint arXiv:1603.04865. https://arxiv.org/abs/1603.04865 (Local File: Paper_19_1603.04865_Robust_Machine_Learning_for_Encrypted_Traffic_Clas.pdf)

2. Abstract (Verbatim / Kept Same): Desktops and laptops can be maliciously exploited to violate privacy. In this paper, we consider the daily battle between the passive attacker who is targeting a specific user against a user that may be adversarial opponent. In this scenario, while the attacker tries to choose the best vector attack by surreptitiously monitoring the victims encrypted network traffic in order to identify users parameters such as the Operating System (OS), browser and apps. The user may use tools such as a Virtual Private Network (VPN) or even change protocols parameters to protect his/her privacy. We provide a large dataset of more than 20,000 examples for this task. We run a comprehensive set of experiments, that achieves high (above 85) classification accuracy, robustness and resilience to changes of features as a function of different network conditions at test time. We also show the effect of a small training set on the accuracy.

3. Introduction: Today, the amount of sensitive data that users handle with devices is growing rapidly. Typical users access their bank web-site, social network web-sites and get information abo ut diseases, sexual or religious preferences that are assumed to be confidential and non-hackable. Personal devices can be attacked in two broad ways which are known as passive and active. Active adversaries try to physically or remotely control the user's device. Passive adversaries may violate the privacy of the user by sniffing the network traffic of the devices from the network side. A passive adversary can use this information to map user actio ns and obtain information on personal interests and habits. If the network traffic is not encrypted, a passive attacker can simply analyze the unencrypted HTTP requests and responses [1]–[4]. However, most Internet traffic today is encrypted a s a result of increased user awareness of privacy threats and encouragement by Google to get all website owners to switch from HTTP to HTTPS by indicating whether sites use secure, encrypted connections in Google ranking algorithms [5], [6]. One immediate outcome has been that traditional Deep Packet Inspection (DPI) methods for information retrieval are no longer viable. Nevertheless, numerous studies have shown that encryption is not sufficient both in desktop and mobile [7]–[30]. Bujlow et al. [23] presented a survey on popular DPI tools for traffic classification. Nguyen and Armitage [24] surveyed machine learning techniques for Internet traffic classification. Mo ore et al. Niemczyk et al. [26] suggested dividing the session into time buckets (10 seconds). They conclude that the new feature (buckets) was almost perfect for Skype but not able t o differentiate between browsers and between joint applicat io

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR) / Cyber Security Research, 2016/2020 | Methodology: Supervised machine learning using burst and flow statistical features over a 20,000+ sample dataset to identify OS, browser, and application inside VPN tunnels. | Key Findings: Achieves >85% classification accuracy even when adversarial users alter protocol cipher parameters or route traffic through encrypted VPN tunnels. | Research Gap: Demonstrates metadata leakage through encrypted VPN tunnels, highlighting the need for automated vulnerability and traffic leakage scoring in IPsec analyzers.] II. RELATED WORKS Feature extraction methods for traffic classification, appl ication classification and application analysis include ses sion duration [36], number of packets in a session [35], [44], minimum, maximum and average values of inter-arrival packets time [35], [36], payload size information [35], bit rate [45], [46], round-trip time [45], packet direction [47], SS L parameters [29] and server sent bit-rate [48]. Liberatore and Levine [49] showed the effectiveness of two traffic analysis techniques for the identification of encryp ted HTTP streams. One is based on a na¨ ■ve Bayes classifier and the other on Jaccard's coefficient similarity measure. They also proposed several methods for actively countering these techniques. They found these methods to be effective, albeit at the cost of a significant increase in the size of the traffic stream. Panchenko et al. [50] showed that a Support V ector Machine (SVM) classifier is able to correctly identif y web pages, even when the user used both encryption and anonymization networks such as Tor [51]. Cai et al. [52] presented a web page fingerprinting attack and showed that it is able to overcome defenses such as the application-leve l defense HTTPOS [53] and randomized pipelining over Tor. Mobile devices, which have different operating systems and different applications implementation leading in many cas es to different network behavior, are also susceptible to atta ckers seeking information on user privacy, applications [54]–[6 0] and actions [8], [61], [62]. Saltaformaggio et al. [61] pres ented NetScope, a passive f

Paper 20 [Pratik Shivekar]: Stealth-MITM DoS Attacks on Secure Channels

1. APA Reference: Herzberg, A., & Shulman, H. (2009). Stealth-MITM DoS Attacks on Secure Channels. arXiv preprint arXiv:0910.3511. https://arxiv.org/abs/0910.3511 (Local File: Paper_20_0910.3511_Stealth_MITM_DoS_Attacks_on_Secure_Channels.pdf)

2. Abstract (Verbatim / Kept Same): We define stealth Man-in-the-Middle adversaries, and analyse their ability to launch denial and degradation of service (DoS) attacks on secure channels. We show realistic attacks, disrupting TCP communication over secure VPNs using IPsec. We present: First amplifying DoS attack on IPsec, when deployed without anti-replay window. First amplifying attack on IPsec, when deployed with a `small' anti-replay window, and analysis of `sufficient' window size. First amplifying attack on IPsec, when deployed with `sufficient' window size. This attack (as the previous) is realistic: attacker needs only to duplicate and speed-up few packets. We also suggest a solution designed to prevent the presented attacks, and to provide secure channel immune to degradation and other DoS attacks. Our solution involves changes (only) to the two gateway machines running IPsec. In addition to their practical importance, our results also raise the challenge of formally defining secure channels immune to DoS and degradation attacks, and providing provably-secure implementations.

3. Introduction: Denial/Degradation of service (DoS) attacks pose an ever growing threat to Internet services and applications. Secure channel protocols, with IPsec [23, 34] being the predominant one, are used to securely connect virtual private networks (VPN), i.e., authenticate data and origin, ensure confidentiality, and prevent replay. IPsec is designed to protect against MITM adversaries that can eavesdrop on the communication and inject spoofed segments into the message stream. It is widely believed, and also specified e.g., in [23], that IPsec also defends higher-layer traffic from DoS attacks when attacker has limited resources (e.g., can only block, inject or reorder a limited number of packets). Defense against DoS attacks is often an important consideration in adopting IPsec for protecting a VPN (rather than say using SSL/TLS [21, 12]). We show that this belief is not precise by presenting several DoS attacks on TCP when used over IPsec. Our attacks raise the following question: what are the properties that secure channel should satisfy to protect against performance degradation attacks? Existing works do not analyse the properties that secure channel protocols should possess to protect against denial of service attacks. There are works that attempt to define what a secure channel is, e.g., [11], but they fail to capture performance analysis of secure channel, i.e., efficiency and resistance to denial of service attacks. Herzberg and Yoffe [20] present a framework that allows to define specifications that capture such properties, and suggest further research on defining secure channel protocols within that framework. However, they do not present such specifications for DoS-preventing secure channel protocols, or demonstrate that existing secure channel protocols fail to prote

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR) / Network Security Study, 2009 | Methodology: Analytical and experimental modeling of Stealth Man-in-the-Middle (MITM) amplifying Denial/Degradation of Service (DoS) attacks against TCP over IPsec ESP. | Key Findings: Proves that packet reordering and duplication by a stealth MITM can exploit IPsec anti-replay window boundaries to severely degrade TCP throughput. | Research Gap: Standard IPsec configurations often leave anti-replay window sizes at default or disabled values; automated configuration auditing is required to prevent DoS amplification.] 2 Model In this section we present the scenario which we consider in the paper, we model and motivate the attacker, and give assumptions on the communication. 2.1 Scenario and Attack Model Consider the scenario presented in Figure 1(a), with a virtual private network between two branches, both connected via gateways GW 1,GW 2 to the Internet. All the communication between the branches is over IPsec, using IPsec's ESP mode with authentication. For simplicity, we assume that the clients are located behind GW 1, and the servers are located behind GW 2. The clients send requests to download files from servers, and the servers send the requested files in response; upon each correctly received data segment, a client generates and send an acknowledgment (ACK). We assume that all communication is over TCP. We consider an attacker located on the Internet between the two gateways, GW 1 and GW 2, in Figure 1(a). The attacker is able to eavesdrop on the communication and inject (a limited number of) packets into the message stream, but cannot drop legitimate packets. Attacker that drops packets can easily disrupt communication and mount a denial of service attack. However, in reality attackers often do not have such abilities. Even when attackers can drop packets they often prefer to refrain when an alternative exists, in order to avoid detection. In addition, we show that the damage inflicted by our stealth attacker is more severe than a damage that a MITM attacker can mount by dropping a limited number of network packets. Note that attacker blocking all communication can clearly moun

Paper 21 [Pratik Shivekar]: Encryption is Futile: Delay Attacks on High-Precision Clock Synchronization

1. APA Reference: Annessi, R., Fabini, J., Iglesias, F., & Zseby, T. (2018). Encryption is Futile: Delay Attacks on High-Precision Clock Synchronization. arXiv preprint arXiv:1811.08569. https://arxiv.org/abs/1811.08569 (Local File: Paper_21_1811.08569_Encryption_is_Futile_Delay_Attacks_on_High_Precisi.pdf)

2. Abstract (Verbatim / Kept Same): Clock synchronization has become essential to modern societies since many critical infrastructures depend on a precise notion of time. This paper analyzes security aspects of high-precision clock synchronization protocols, particularly their alleged protection against delay attacks when clock synchronization traffic is encrypted using standard network security protocols such as IPsec, MACsec, or TLS. We use the Precision Time Protocol (PTP), the most widely used protocol for high-precision clock synchronization, to demonstrate that statistical traffic analysis can identify properties that support selective message delay attacks even for encrypted traffic. We furthermore identify a fundamental conflict in secure clock synchronization between the need of deterministic traffic to improve precision and the need to obfuscate traffic in order to mitigate delay attacks. A theoretical analysis of clock synchronization protocols isolates the characteristics that make these protocols vulnerable to delay attacks and argues that such attacks cannot be prevented entirely but only be mitigated. Knowledge of the underlying communication network in terms of one-way delays and knowledge on physical constraints of these networks can help to compute guaranteed maximum bounds for slave clock offsets. These bounds are essential for detecting delay attacks and minimizing their impact. In the general case, however, the precision that can be guaranteed in adversarial settings is orders of magnitude lower than required for high-precision clock synchronization in critical infrastructures, which, therefore, must not rely on a precise notion of time when using untrusted networks.

3. Introduction: C LOCK synchronization protocols have become an essential building block of numerous applications that rely on a precise notion of time. The deployment of clock synchronization for controlling system clocks of critical applications in telecommunication, industrial automation, financial markets, avionics, or energy distribution has increased the dependency of critical infrastructures on clocks synchronized with increasingly high precision. Many processes, especially in measurement, control, telecommunications, industrial and financial applications are not only sensitive to errors in the value domain but also to errors in the time domain. Examples for strict dependencies on precise time are safety-critical applications in the Smart Grid, which require an accuracy of 1 to 100 µs (10 µs in case of current differential line protection with high fault current sensitivity [1, 2]) or MiFID II in the financial sector, requiring an accuracy of up to 100 µs [3, 4, 5]. Cellular networks also have strong requirements for synchronized clocks (≤ 1 µs) [6]. Errors in clock synchronization can lead to wrong timings and may therefore originate faulty sensor reports, endanger control decisions, and adversely affect the overall functionality of a wide range of (critical) services that depend on accurate time. In recent years, security of clock synchronization received increased attention as various attacks on clock synchronization protocols (and countermeasures) were proposed. For this reason, clock synchronization protocols need to be secured whenever used outside of fully trusted network environments. Clock synchronization protocols are specifically susceptible to delay attacks since the times when messages are sent and received have an actual effect on the receiver’s notion of time. Del

4. Literature Review & Findings: [Venue/Publisher: IEEE Transactions / arXiv Cryptography and Security (cs.CR), 2018 | Methodology: Statistical traffic analysis (packet length, timing, direction) and selective delay attacks against Precision Time Protocol (PTP) encapsulated inside IPsec tunnels. | Key Findings: Demonstrates that IPsec encryption alone does not prevent statistical identification of encapsulated time-sync messages or selective delay attacks. | Research Gap: Proves that protocol analyzers must evaluate both cryptographic configuration (ciphers/modes) and statistical flow side-channels.] V. RELATED WORK Tsang and Beznosov where the first to analyze the security of PTP in 2005 [15, 16]. They describe delay attacks briefly and propose to average delay measurements as a countermeasure. We argue that averaging delay measurements also averages the malicious delay such that the attack is only barely mitigated. The attack has even full effect as soon as the duration of the attack is longer than the averaging interval 5. The 4Sometimes, asymmetry is even intentional like with Ethernet where cables are asymmetric by design to reduce far end crosstalk [14]. 5And if the averaging interval is too long, the clock synchronization precision may be negatively affected (since reacting on changes in the RTD would take too long). Fig. 3. PTP clock synchronization with delayed SYNC message. authors furthermore propose to check for abnormal values. But abnormal values (i.e., spikes in the measured clock offset) only occur for simple delay attacks but not for incremental delay attacks (as to be shown in Section VI-C). PTP version 2 includes an experimental security extension, Annex K [11], which provides message integrity and replay protection. In Annex K, however, several flaws were discovered and it was never properly formalized [17, 18]. In any case, Annex K does not cover attacks in the time domain at all. Interestingly, the PTP standard [11] mentions link asymmetry (although in a single sentence only) but does not cover delay attacks any further. We analyze asymmetric link delay attacks in Section VIII and countermeasures against such attacks in Section IX. Network Time Sec

Paper 22 [Pratik Shivekar]: Quantum Cryptography in Practice

1. APA Reference: Elliott, C., Pearson, D., & Troxel, G. (2003). Quantum Cryptography in Practice. arXiv preprint arXiv:quant-ph/0307049. https://arxiv.org/abs/quant-ph/0307049 (Local File: Paper_22_quant-ph_0307049_Quantum_Cryptography_in_Practice.pdf)

2. Abstract (Verbatim / Kept Same): BBN, Harvard, and Boston University are building the DARPA Quantum Network, the world's first network that delivers end-to-end network security via high-speed Quantum Key Distribution, and testing that Network against sophisticated eavesdropping attacks. The first network link has been up and steadily operational in our laboratory since December 2002. It provides a Virtual Private Network between private enclaves, with user traffic protected by a weak-coherent implementation of quantum cryptography. This prototype is suitable for deployment in metro-size areas via standard telecom (dark) fiber. In this paper, we introduce quantum cryptography, discuss its relation to modern secure networks, and describe its unusual physical layer, its specialized quantum cryptographic protocol suite (quite interesting in its own right), and our extensions to IPsec to integrate it with quantum cryptography.

3. Introduction: Modern networks generally rely on one of two basic cryptographic techniques to ensure the confidentiality and integrity of traffic carried across the network : symmetric (secret) key and asymmetric (public) key. Indeed today's best systems generally employ both, using public key system s for authentication and establishment of secret " session" keys, and then protecting all or part of a traffic flow with these session keys. Certain other systems transport secret keys "out of channel," e.g. via courier, as in classical cryptography. Fundamental aspects of quantum physics – unitarity, the uncertainty principle, and the Einstein -Podolsky-Rosen violation of Bell's inequalities – now suggest a third paradigm for key distribution: quantum cryptography. In itial experiments seem to confirm the utility of this paradigm. Assuming that the theoretical models continue to be confirmed in the use of actual devices, the fundamental laws of nature can be invoked to assure the confidentiality of transmitted data. As shown in Fig. 1, quantum cryptography – more properly termed Quantum Key Distribution, QKD – employs two distinct channels. One is used for transmission of quantum key material by very dim (single photon) light pulses. The other, public channel carr ies all message traffic, including the cryptographic protocols, encrypted user traffic, etc. QKD consists of the transmission of raw key material, e.g., as dim pulses of light from Alice to Bob, via the quantum channel, plus processing of this raw ma terial to derive the actual keys. This processing involves public communication (key agreement protocols) between Alice and Bob, conducted in the public channel, along with specialized QKD algorithms. The resulting keys can then be used for crypto graphic purposes, e.g., to protect

4. Literature Review & Findings: [Venue/Publisher: ACM SIGCOMM 2003 Conference Proceedings, ACM | Methodology: Design and metro-fiber implementation of the DARPA Quantum Network integrating BB84 QKD high-speed key delivery with IPsec VPNs and IKE extensions. | Key Findings: Established the foundational architecture for rapid-reseeding IPsec Security Associations using out-of-band quantum key material. | Research Gap: Early foundational work using IKEv1 extensions; modern deployments require IKEv2 (RFC 7296/9370) and automated security posture verification.] 2. DESIRABLE QKD ATTRIBUTES Broadly stated, QKD offers a technique for coming to agreement upon a shared random sequence of bits within two distinct devices, with a very low probability that other devices (eavesdroppers) will be able to make successful inferences as to those bits' values. In specific practice, such sequences are then used as secret keys for encoding and decoding messages between the two devices. Viewed in this light, QKD is quite clearly a key distribution technique, and one can rate QKD's strengths against a number of important goals for key distribution, as summarized in the following paragraphs. Confidentiality of Keys. Confidentiality is the main reason for interest in QKD. Pu blic key systems suffer from an ongoing uncertainty that decryption is mathematically intractable. Thus key agreement primitives widely used in today's Internet security architecture, e.g., Diffie -Hellman, may perhaps be broken at some point in the future . This would not only hinder future ability to communicate but could reveal past traffic . Classic secret key systems have suffered from different problems, namely, insider threats and the logistical burden of distributing keying material. Assu ming that QKD techniques are properly embedded into an overall secure system, they can provide automatic distribution of keys that may offer security superior to that of its competitors. Authentication. QKD does not in itself provide authentication. Current strategies for authentication in QKD systems include prepositioning of secret keys at pairs of devices, to be used in hash-based auth

Paper 23 [Pratik Shivekar]: A method to Implement the Kerberos User Authentication and the secured Internet Service

1. APA Reference: Song-Ho, P., Myong-Suk, P., & Chung-Hyok, J. (2016). A method to Implement the Kerberos User Authentication and the secured Internet Service. arXiv preprint arXiv:1604.08799. https://arxiv.org/abs/1604.08799 (Local File: Paper_23_1604.08799_A_method_to_Implement_the_Kerberos_User_Authentica.pdf)

2. Abstract (Verbatim / Kept Same): This paper proposes a PKINIT_AS Kerberos V5 authentication system to use public key cryptography and a method to implement the gssapi_krb authentication method and secured Internet service using it in IPSec VPN

3. Introduction: Kerberos is a network authentication protocol created by MIT, and uses symmetric-key cryptography to authenticate users to network services. And IPSec is a typical security protocol that is used to construct VPN (virtual private network) in network layer of TCP/IP protocol. When implementing secured Internet service combining Kerberos with IPsec security program, there are some problems. ① Using public key cryptography in Kerberos authentication system Kerberos that was proposed as authentication system based on trusted third-party have been now developing by version 5[1] and it used public key cryptography to verify client's identification and share a secret key between user and Authentication Server or TGS server or Application Server[2, 4]. 2 Specially, the method to use ECDH key exchange and ECSig signature in sharing secret key between the user and the AS server [5] is proposed, but program to implement these methods completely is not yet published. ② Implementing gssapi_krb user authentication in IPSec Because the combination parameters between IPSec and Kerberos are generally set, the gssapi_krb authentication method[3,6,7] that the present IPSec proposes can be now not used. These parameters must be set according to application service name and mechanism type, etc. ③ Implementing and using of secured Internet service in Kerberos-aware applications Httpd daemon with Kerberos authenticati on module(mod_auth_kerb) and Squid program with Kerberos authentication module(negotiate _auth) offer only a se cured www service, and Kerberos V5 FTP program offers only secu red ftp service of command-driven mode. (Kerberos-aware Email application is also same as Kerberos V5 FTP program) In the former case, user must configure Firefox browser to use Kerberos for Single Sign-on.

4. Literature Review & Findings: [Venue/Publisher: arXiv Cryptography and Security (cs.CR), 2016 | Methodology: Integration of Public Key Cryptography for Initial Authentication in Kerberos V5 (PKINIT_AS with ECDH/ECSig) and GSSAPI (`gssapi_krb`) into IPsec VPNs. | Key Findings: Enables centralized enterprise Single Sign-On (SSO) and mutual authentication for IPsec VPN clients using Kerberos tickets and elliptic-curve signatures. | Research Gap: Complex GSSAPI-IPsec parameter bindings are prone to misconfiguration and require automated verification tools.] C 3 Where, Ticket TGS :{Flags,KC,TGS,RealmC,IDC,ADC,Times}KTGS ③ C->TGS:Options,IDS,Times,Nonce2,TicketTGS,AuthenticatorC Where, AuthenticatorC : {IDC,RealmC,TS1}KC,TGS ④ TGS->C:RealmC,IDC,TicketS,{KC,S,Times,Nonce2,RealmS,IDS} KC,TGS Where, TicketS : {Flags,KC,S,RealmC,IDC,ADC,Times}KS ⑤ C->S:Options,TicketS,AuthenticatorC Where, Authenticator C: {IDC,RealmC,TS1}KC,S ⑥ S->C :{TS2,Subkey,seq#}KC,S Where, Pc indicates public key of client C, PC-1 indicates private key of client C. The authentication process of PKINIT_AS Kerber os authentication system proposed in this paper proceeds as follows. ① Client C sends signing its identification as own privat e key and its own public key certificate beside default request message to the AS. ② When AS receives the request message from c lient C and processes it, it identifies that public key of public key certificate in received message is the sa me as public key of public key certificate for the client in database, if these publi c keys are the same, it verifies signature using the key. If signature verification is su ccessful, it proceeds with next step. AS sends encrypting secret session key(kc.tgs) between client C and Ticket Granting Server(TGS) with client's public key to the client. ③ Client decrypts received reply message from AS with secret Kc created by its password and then decrypts encrypted k c.tgs with its own private key P C-1 to get secrete session key k c.tgs between client and TGS. The rest reception processes in client and the next steps of Ke rberos authentication system are also same as default Kerberos processes

Paper 24 [Pratik Shivekar]: Navigating Quantum Security Risks in Networked Environments: A Comprehensive Study of Quantum-Safe Network Protocols

1. APA Reference: Baseri, Y., Chouhan, V., & Hafid, A. (2024). Navigating Quantum Security Risks in Networked Environments: A Comprehensive Study of Quantum-Safe Network Protocols. Computers & Security, Volume 142, July 2024, 103883. https://doi.org/10.1016/j.cose.2024.103883 (arXiv:2404.08232) (Local File: Paper_24_2404.08232_Navigating_Quantum_Security_Risks_in_Networked_Env.pdf)

2. Abstract (Verbatim / Kept Same): The emergence of quantum computing poses a formidable security challenge to network protocols traditionally safeguarded by classical cryptographic algorithms. This paper provides an exhaustive analysis of vulnerabilities introduced by quantum computing in a diverse array of widely utilized security protocols across the layers of the TCP/IP model, including TLS, IPsec, SSH, PGP, and more. Our investigation focuses on precisely identifying vulnerabilities susceptible to exploitation by quantum adversaries at various migration stages for each protocol while also assessing the associated risks and consequences for secure communication. We delve deep into the impact of quantum computing on each protocol, emphasizing potential threats posed by quantum attacks and scrutinizing the effectiveness of post-quantum cryptographic solutions. Through carefully evaluating vulnerabilities and risks that network protocols face in the post-quantum era, this study provides invaluable insights to guide the

development of appropriate countermeasures. Our findings contribute to a broader comprehension of quantum computing's influence on network security and offer practical guidance for protocol designers, implementers, and policymakers in addressing the challenges stemming from the advancement of quantum computing. This comprehensive study is a crucial step toward fortifying the security of networked environments in the quantum age.

3. Introduction: Quantum computing has emerged as a powerful field that combines principles from physics and computer science. It offers the potential to solve complex problems at an astonishing speed, revolutionizing various domains. However, this remarkable computational power raises concerns about the security of network protocols that rely on traditional cryptographic algorithms. Quantum computers have the ability to break widely-used encryption algorithms, such as RSA and Elliptic Curve Cryptography (ECC), which are the foundations of secure communication in network protocols (Shor, 2022 ; Grover , 2023). In organizations, security protocols are extensively employed to verify the source and protect the confidentiality and integrity of information during communication and storage. Current security protocols, such as TLS and SSH, are effective against classical computer attacks on network communications (Portmann and Renner, 2022). However, the emergence of fault-tolerant quantum computers could pose a significant threat to the security of these protocols by exploiting the underlying mathematical challenges in a matter of hours or even seconds (Preskill, 2018). ∗ Corresponding author. E-mail addresses: yaser.baseri@umontreal.ca (Y. Baseri), vikas.chouhan@unb.ca (V. Chouhan), ahafid@iro.umontreal.ca (A. Hafid). The growing impact of quantum computing on network protocols has recently drawn significant attention. As quantum computers continue to advance, their ability to breach traditional cryptographic algorithms raises serious concerns regarding network communication security. In response to this challenge, researchers have been actively exploring post-quantum cryptographic solutions capable of withstanding quantum computer attacks. However, the crucial step remains evaluating the

4. Literature Review & Findings: [Venue/Publisher: Computers & Security, Vol. 142, 103883 (2024), Elsevier | Methodology: Comprehensive multi-layer vulnerability taxonomy of TCP/IP security protocols (IPsec, IKEv2, TLS, SSH, PGP) across quantum migration stages. | Key Findings: Maps specific cryptographic vulnerabilities in IPsec AH, ESP, and IKEv2 against Shor's and Grover's algorithms and evaluates hybrid migration risks. | Research Gap: Provides a theoretical vulnerability matrix that needs to be operationalized into an automated AI-powered IPsec security assessment tool.] 2. Related works The emergence of quantum computing has sparked significant research into the vulnerabilities of network protocols within the security landscape. This section provides an overview of prior research in this domain, encompassing foundational work and recent contributions. 2.1. Quantum computing and cryptanalysis Quantum computing’s theoretical foundations and their implications for traditional cryptographic algorithms have been rigorously examined. Of note, Shor’s algorithm’s ability to efficiently factor large numbers, potentially compromising widely-used public-key encryption methods like RSA, has garnered significant attention (Shor, 1994). Additionally, Grover’s algorithm, with its unstructured search capabilities that threaten symmetric key encryption, has been explored (Grover, 1996). Recent studies continue to deepen our understanding of quantum computing’s vulnerabilities (Ji et al., 2022). 2.2. Post-quantum cryptography The emerging area of post-quantum cryptography focuses on developing cryptographic solutions capable of withstanding quantum attacks. Recent notable contributions encompass lattice-based cryptography, code-based cryptography, hash-based, and isogeny-based schemes (Ding et al., 2017; Lyubashevsky, 2016). The NIST PostQuantum Cryptography Standardization project has played a central role in advancing this research, providing a framework for evaluating and selecting post-quantum cryptographic algorithms (National Institute of Standards and Technology (NIST), 2023b). These endeavors pave the way for exploring alternative cryptographic meth

Paper 25 [Pratik Shivekar]: Quantum Key Distribution (QKD) and Commodity Security Protocols: Introduction and Integration

1. APA Reference: Mink, A., Frankel, S., & Perlner, R. (2010). Quantum Key Distribution (QKD) and Commodity Security Protocols: Introduction and Integration. International Journal of Network Security & Its Applications 1.2 (2009) 101-112. https://arxiv.org/abs/1004.0605 (Local File: Paper_25_1004.0605_Quantum_Key_Distribution_QKD_and_Commodity_Securit.pdf)

2. Abstract (Verbatim / Kept Same): We present an overview of quantum key distribution (QKD), a secure key exchange method based on the quantum laws of physics rather than computational complexity. We also provide an overview of the two most widely used commodity security protocols, IPsec and TLS. Pursuing a key exchange model, we propose how QKD could be integrated into these security applications. For such a QKD integration we propose a support layer that provides a set of common QKD services between the QKD protocol and the security applications

3. Introduction: Quantum Key Distribution (QKD) [1, 2] is a technology, based on the quantum laws of physics, rather than the assumed computational complexity of mathematical problems, to generate and distribute provably secure

cipher keys over unsecured channels. It does this using single photon technology and can detect pote ntial eavesdropping via the quantum bit error rates of the quantum channel. Sending randomly encoded information on single photons produces a shared secret that is a random string and the probabilistic nature of measuring the photon state provides the basis of its security. A QKD system consists of a quantum channel and a classical channel. The quantum channel is only used to transmit qbits (single photons) and must consist of a transparent optical path (fiber, free-space and optical switches, no routers, amplifiers or copper). It is a lossy and probabilistic channel. The classical channel can be a conventional IP channel (not necessarily optical), but depending on system design it may need to be dedicated and closely tied to the quantum channel for timing requirements. It would not be unusual for the quantum and classical channels to share a common fiber via wavelength division multiplexing (WDM). A quantum network connects a number of point-to-point QKD systems together so that one can develop shared secrets between users anywhere on that sub-network. A quantum network would be an embedded sub-network within a conventional communication network for the purpose of developing shared secrets, not transporting secure messages. The physical link (e.g., fiber) that carries the classical channel could certainly support general messages, but not within the QKD classical channel. Commodity security protocols, such as Internet Protocol Security (IPsec) and Transport Layer

4. Literature Review & Findings: [Venue/Publisher: International Journal of Network Security & Its Applications (IJNSA), Vol. 1, No. 2 (NIST) | Methodology: Architectural design by NIST researchers for a common key-management support layer integrating QKD services with commodity IPsec and TLS stacks. | Key Findings: Defines standardized API abstractions to decouple key generation from IPsec IKE/ESP state machines while preserving Perfect Forward Secrecy (PFS). | Research Gap: Highlights the lack of standardized cross-vendor diagnostic APIs across IPsec implementations, motivating non-intrusive protocol analysis.] 2. QKD Overview The BB84 [7] protocol and its variants are the only known provably secure QKD protocols. Other QKD protocols (e.g., differ ential phase shift keying [8]), alth ough promising, have yet to be proven secure. The BB84 protocol consists of four stages (See Fig 1). The first stage is the transmission of the randomly encoded single photon stream over the quantum channel from Alice (the sender) to Bob (the receiver) to establish the initial raw key. Alice maintains a temporary database of the state of each photon sent. The second stage is sifting, where Bob sends a list of photons detected and their basis, but not their value, back to Alice over the classical channel. Basis refers to how the photons were measured. Photons can be encoded in one of two bases (e.g., horizontal/vertical or di agonal polarization). There is only one photon and it can only be measured once, so only one basis can be applied. If it’s measured in the correct basis the value measured will be correct. If it’s measured in th e wrong basis, the value will be random. Alice retains, from its databa se, only those entries received by Bob in the correct basis and sends this revised list back to Bob over the classical channel. Bob retains only those entries on this revised list. Alice and Bob now have a list of sifted keys. These lists are of the same length but may have some errors between them. This is the quantum bit error rate and it is an indication of eavesdropping. The third stage is reconciliation to correct these errors. Cascade [9, 10] and its variants are the predom inant reconciliation a

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